



R2_DEPI2_CAI2_ISS4_S1_Cisco Network Administrator Spring semester

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Proposal for

Developing a Smart Office Network Solution

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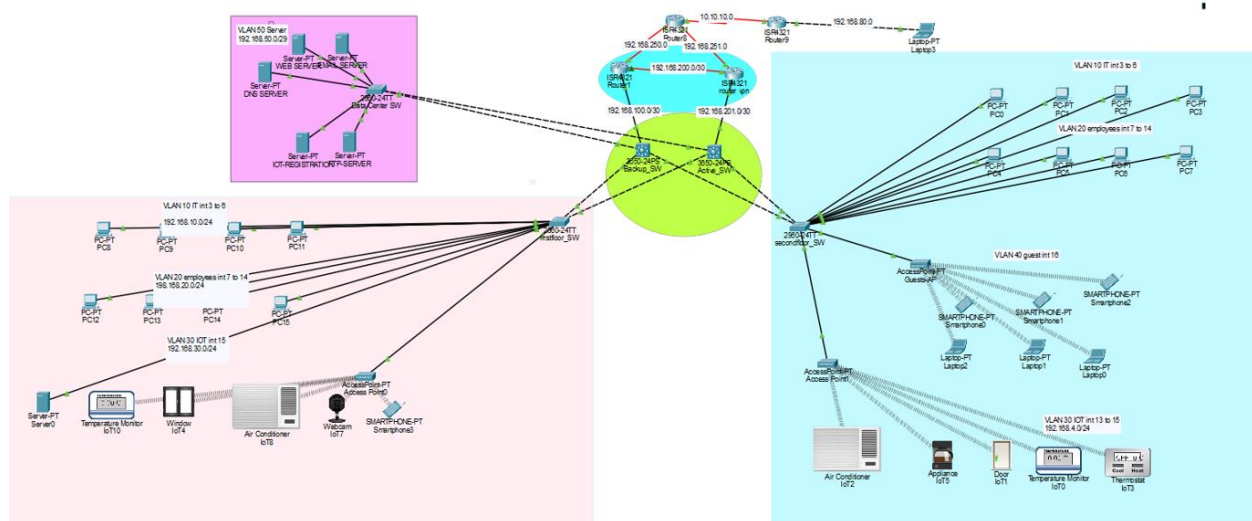
Network Engineering Students Group code: CAI2_ISS4_S1

1. Project Overview

This project presents a secure and scalable Smart Office Network solution for ByteWave Technologies, a software development company. The goal of the network design is to:

- Ensure high performance and reliable connectivity.
- Support IoT integration, including smart lighting, climate control, and security systems.
- Enhance employee productivity, collaboration, and comfort.
- Provide robust security features such as VPN access and a Zero Trust model.
- Allow remote monitoring and support future scalability.

2. Network Topology



3. Topology Components

- Two Layer 3 Core Switches (HSRP-enabled)
- Multiple Layer 2 Access Switches
- DHCP Server
- Wireless Access Point
- ISP Router
- IoT Devices
- Employee and Guest Devices

4. VLANs

The network is segmented using VLANs to improve security and performance:

- VLAN 10: Admin
- VLAN 20: Developers
- VLAN 30: IoT
- VLAN 40: Guest (Wireless)
- VLAN 50: Data center

5. Protocols & Technologies Used

- VLAN
- DHCP
- HSRP
- OSPF
- NAT
- VPN
- SSH / Telnet
- Wireless Security

6. Configuration steps

• VLAN Configuration (Layer 2 Switch)

On Layer 2 switches, we created VLANs to logically segment the network and enhance performance, security, and management. Each VLAN represents a specific department or device group:

1. VLAN 10 – Admin: Created for administrative staff to isolate sensitive communication and management traffic.
2. VLAN 20 – Developers: Assigned to developers to provide them with a dedicated, high-bandwidth working environment.
3. VLAN 30 – IoT Devices: Used to group all IoT devices under one broadcast domain, ensuring they're isolated from other departments for security and control.
4. VLAN 40 – Guests: Reserved for wireless guest access to separate visitors from the internal network, protecting core resources.
5. This configuration reduces broadcast domains, enhances security, and improves traffic management

• VLAN Configuration Verification

We used the command `show vlan brief` to verify that all VLANs were correctly created and assigned. This step ensures no mistakes were made during configuration and helps us confirm port-to-VLAN mappings.

• VLAN Configuration (Layer 3 Switch)

On the Layer 3 switch, we created Switched Virtual Interfaces (SVIs) for each VLAN to enable inter-VLAN routing. Without these, devices in different VLANs wouldn't be able to communicate.

Each interface was assigned a default gateway for its VLAN:

1. interface vlan10 with IP 192.168.10.1 for Admin
2. interface vlan20 with IP 192.168.20.1 for Developers
3. interface vlan30 with IP 192.168.30.1 for IoT
4. interface vlan40 with IP 192.168.40.1 for Guests

This allows Layer 3 routing between VLANs internally within the smart office.

- **DHCP Configuration**

Instead of manually assigning IP addresses, we enabled a DHCP server on the Layer 3 switch to dynamically assign IPs to clients in each VLAN.

For example:

1. VLAN 10 devices received IPs from the 192.168.10.0/24 pool.
2. VLAN 20 devices received IPs from the 192.168.20.0/24 pool.

This automated IP assignment ensures efficient address management and avoids conflicts

- **DHCP Connectivity Verification**

The command `show ip dhcp binding` was used to verify which devices obtained IP addresses. It confirms DHCP is functioning properly and that devices are receiving correct IPs from the right pools.

- **HSRP Implementation**

We used Hot Standby Router Protocol (HSRP) to provide gateway redundancy. Two Layer 3 switches share a virtual IP (e.g., 192.168.10.254), acting as a single default gateway for clients.

1. One switch is the active gateway, handling traffic.
2. The second is in standby mode and takes over if the active one fails.

This ensures high availability and network reliability, preventing outages in case a switch goes down

- **OSPF Routing Protocol**

We used OSPF (Open Shortest Path First) as our interior gateway protocol to dynamically share routing information between Layer 3 switches.

1. It calculates the most efficient path between devices.
2. It adapts quickly to network changes or failures.

This is ideal for a scalable, enterprise-grade environment like our smart office.

- **IoT Devices Control**

We built a dedicated server in VLAN 30 to manage and control IoT devices such as smart lights, sensors, and climate control systems. This server:

1. Acts as a DHCP server for IoT devices.
2. Hosts the centralized control system, accessible via smartphone.
3. Ensures IoT traffic is isolated from business and guest traffic for security and performance.

This setup enables real-time monitoring and control through a mobile app.

- **NAT Overload**

To allow devices inside the private VLANs to access the internet, we used NAT Overload (PAT):

1. Translates multiple internal private IPs into one public IP.
2. Provides internet access to internal users.
3. Maintains security by hiding internal IPs from the outside.

This is essential for accessing online tools, cloud platforms, and remote updates

- **Wireless VLAN Access**

We created a Guest VLAN (VLAN 40) for wireless users. It is configured with a separate SSID (Guests) and password (AP123). This:

1. Offers internet access to visitors.
2. Ensures they are completely isolated from internal company resources.
3. Helps maintain network security and bandwidth management

- **VPN Connectivity**

We implemented a Site-to-Site VPN between our office network and a remote laptop through the ISP router. This VPN:

1. Establishes a secure encrypted tunnel between two networks.

2. Allows secure access to internal systems from remote locations.
3. Does not require client software, unlike remote-access VPNs.

It's perfect for IT administrators or employees working remotely who need constant access to internal services.

- **SSH / Telnet Configuration**

We configured remote management access via SSH and Telnet on the switches:

1. SSH provides secure encrypted access to manage devices.
2. Telnet was also enabled for basic legacy support.
3. Local authentication was set to prevent unauthorized access.

This ensures network administrators can safely manage infrastructure remotely.

Conclusion

This Smart Office Network successfully integrates security, scalability, IoT support, and reliable high-speed connectivity. The implemented technologies and configurations ensure that ByteWave Technologies can operate efficiently while being future-ready.